



CHAPTER I – INTRODUCTION



SECTION: A. Background of the Study



What is it?

The background introduces the **topic** of the project, **explains the current situation or problem**, and presents the **proposed solution** (your system). It tells the reader **why the study matters**.



Steps to Write:

1. **Start with a general topic** (e.g., what is inventory? Why is it important?)
2. **Focus on your institution/program** (e.g., ISAT U Dumangas, BIT Automotive)
3. **Describe the current situation or problem** (manual process, inefficiencies)
4. **Explain the gap or need for improvement** (tool loss, no real-time tracking)
5. **Present your solution** (the proposed system)
6. **End with expected benefits** (better tracking, CHED compliance, reduced workload)



Example:

Inventory systems are essential tools used in schools, companies, and organizations to monitor the availability, condition, and location of equipment and supplies. In educational institutions, particularly those offering technical and vocational programs, proper inventory management ensures that tools needed for laboratory activities are always available and in good condition.

At Iloilo Science and Technology University Dumangas Campus, the Bachelor of Industrial Technology (BIT) major in Automotive Technology includes laboratory courses that require the use of various tools and equipment. However, the Automotive Department currently uses a **manual inventory system**. All tool transactions are recorded in notebooks or spreadsheets. This manual process is time-consuming, prone to errors, and difficult to monitor in real-time.

There have been instances where tools were not returned, misplaced, or damaged, but no clear record could verify the borrower or transaction details. This affects laboratory efficiency and accountability. Moreover, generating reports for tool usage and stock status takes too much time.

Therefore, this study proposes the development of a **Tools Inventory System** that will automate and streamline the inventory process. This system will allow the department to record borrowing and returning transactions, monitor stock-in and stock-out activities, manage

user accounts, and generate real-time reports. It aims to improve accountability, reduce tool loss, and support effective teaching and learning.

SECTION: B. Objectives of the Study

What is it?

This section explains what your **system will accomplish**. It is divided into:

- **General Objective** (overall goal)
- **Specific Objectives** (step-by-step functions)

Steps to Write:

1. Write 1 **general objective**: what the system as a whole aims to achieve.
2. Write **specific objectives** using action verbs (e.g., develop, manage, generate, evaluate).
3. Each objective should refer to one core function of your system.

Example:

General Objective:

This study aims to develop a **Tools Inventory System** for the BIT Automotive Program of ISAT U Dumangas Campus.

Specifically, it aims to:

1. create a database of tools and equipment used in the laboratory;
2. manage user accounts for administrators and laboratory staff;
3. manage the registry of tools and equipment;
4. record and monitor stock-in and stock-out transactions;
5. record and monitor tool borrowing and returning transactions;
6. generate inventory reports; and
7. evaluate the system using the ISO 25010 software quality standards.

SECTION: C. Conceptual Framework

What is it?

This part shows how the **system works logically**, using the **IPO (Input-Process-Output)** model.

Steps to Write:

1. Identify your **Inputs**: the data your system will receive.
2. Identify the **Processes**: what the system will do with the data.
3. Identify the **Outputs**: what the system will generate or produce.
4. Create a **diagram** showing IPO flow (use draw.io or PowerPoint).
5. Describe the framework in paragraph form.

Example Presentation and Description:

Figure 1 shows the conceptual framework of this study.

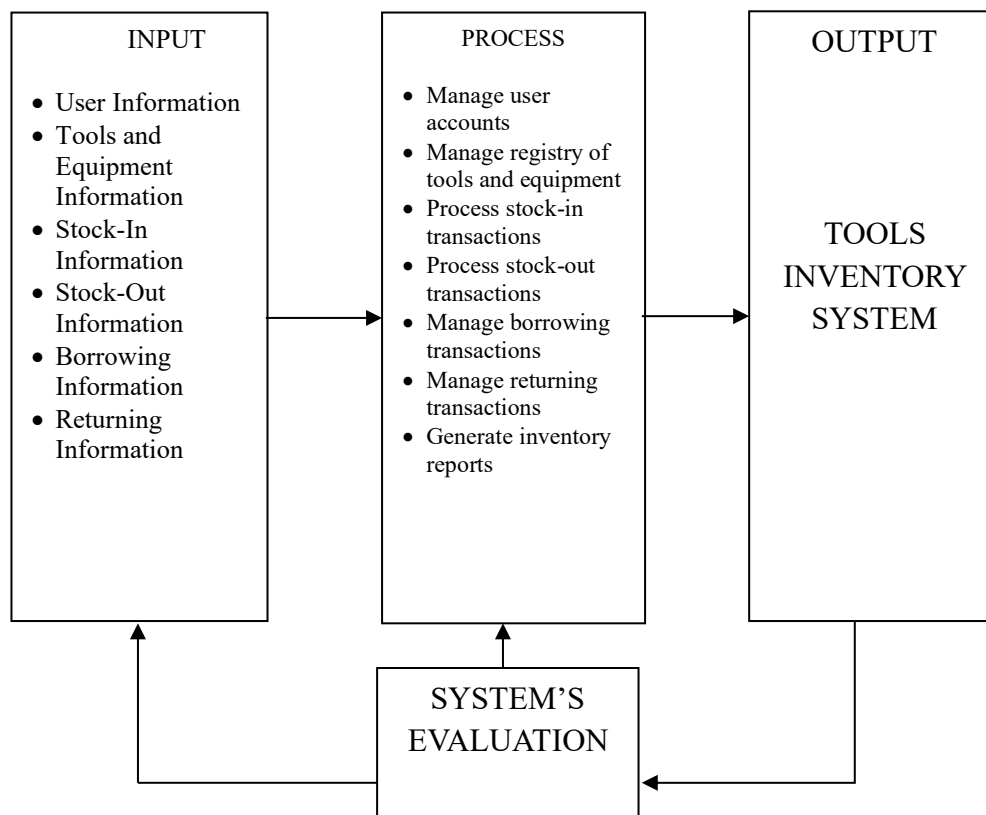


Figure 1. *Conceptual Framework of the Study*

The conceptual framework of the study, presented in Figure 1, follows the Input-Process-Output (IPO) model, which outlines the flow of activities involved in developing the Tools Inventory System for the BIT Automotive Program of Iloilo Science and Technology University Dumangas Campus. The input stage consists of the essential data and information required by the system, including user information, tools and equipment details, stock-in and stock-out records, borrowing information, and returning information. These inputs serve as the foundation for the system's operations and ensure that processes are based on complete and accurate data.

The process stage involves the core activities needed to transform the inputs into the desired system output. These include managing user accounts, maintaining the registry of tools and equipment, processing stock-in and stock-out transactions, managing borrowing and returning transactions, and generating inventory reports. These processes are designed to ensure that the system operates efficiently, provides accurate records, and supports the needs of laboratory operations.

The output of the framework is the Tools Inventory System, a functional system that improves the management of tools and equipment in the Automotive Department. To ensure the system's quality and effectiveness, the framework includes a system evaluation phase, where the developed system is assessed using the ISO 25010 software quality model. This evaluation examines the system's functionality, reliability, usability, and maintainability. The results of the evaluation serve as feedback, guiding further improvements by refining inputs and processes to better meet the needs of the department.

SECTION: D. Definition of Terms

What is this section about?

This section defines all important **technical**, **system**, or **domain-specific** words found in your capstone. Each term should be explained using:

- A **Conceptual Definition** (from books or journals),
 - An **Operational Definition** (your own use-case).
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Steps to Create This Section:

1. List **terms** related to your system.
 2. Search for **academic definitions** or use dictionary sources.
 3. Start each entry like this:
 - **Term** – *Conceptual Definition* (with citation).
 - *In this study, [Operational Definition]*.
 4. Keep it consistent in format and tone.
 5. Arrange terms in **alphabetical order**.
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FULL EXAMPLE WITH CLEAR LABELING:

Borrowing Transactions

Conceptual Definition:

The act of temporarily acquiring tools, equipment, or resources with the responsibility to return them. (Bertland, 2017)

Operational Definition:

In this study, borrowing transactions refer to the **system function** that records the date, borrower details, and quantity of tools borrowed by students or staff in the automotive laboratory.

Database

Conceptual Definition:

A structured collection of data stored electronically, allowing easy access, retrieval, and management using a database management system. (Coronel & Morris, 2016)

Operational Definition:

In this study, the database refers to the **MySQL-powered backend** that stores user information, inventory records, stock transactions, and reports within the Tools Inventory System.

Stock-In

Conceptual Definition:

The process of adding items or materials into the inventory system for future use. (Stevenson, 2018)

Operational Definition:

In this study, stock-in refers to the module that records **new tools or restocked items** being added to the laboratory inventory through the system.

Tools Inventory System

Conceptual Definition:

A software platform used to monitor, record, and manage tools and equipment to ensure availability and proper usage. (Laudon & Laudon, 2020)

Operational Definition:

In this study, the Tools Inventory System refers to the **developed LAN-based application** that allows the BIT Automotive Program to automate tool monitoring, borrowing, and reporting.

User Accounts

Conceptual Definition:

Personalized login credentials used to gain authorized access to a digital system with varying roles and permissions. (Stallings, 2017)

Operational Definition:

In this study, user accounts refer to the **admin and laboratory staff profiles** that log in to the system to perform tasks such as managing tools and viewing reports.

Stock-Out

Conceptual Definition:

A condition where inventory levels of a specific item become zero, making the item temporarily unavailable. (Hendricks & Singhal, 2005)

Operational Definition:

In this study, stock-out refers to the **system's feature** for logging tools that are disposed of, transferred, or lost from the inventory.

Tip:

Always cite your conceptual definitions properly (APA style), and always begin your operational definition with “**In this study...**”

SECTION: Significance of the Study

Instruction:

Explain why this system is important for each stakeholder.

Sample Content:

Faculty – Allows automated tracking of tools, minimizing manual errors.

Laboratory Staff – Simplifies tool issuance and return processes.

Students – Ensures tools are available for their lab activities.

Program Coordinator – Helps monitor tool utilization and inventory needs.

University – Enhances transparency and CHED compliance.

Future Researchers – Serves as reference for similar systems using ISO 25010 standards.

SECTION: F. Scope and Limitations

What is “Scope and Limitations” in a Capstone Project?

This section is a **critical component** of Chapter I. It outlines:

- **SCOPE** – What your system **includes**: features, users, environment, modules, technologies.
- **LIMITATIONS** – What your system **does not include**: boundaries, excluded features, constraints due to time, tools, or design decisions.

It defines the **boundary** of your project clearly and sets **realistic expectations** for the readers, clients, and panel members.

Why is this Section Important?

- ✓ It tells readers **what to expect** from your system and what not to expect.
 - ✓ It shows that the project is **focused, realistic, and feasible**.
 - ✓ It protects you from being criticized for features or goals you never intended to include.
 - ✓ It reveals a **professional level of planning** and **self-awareness** about project limitations.
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HOW TO WRITE THIS SECTION

Part 1: Scope (What the system covers)

Clearly describe:

- **Main system modules** (e.g., borrow/return, inventory tracking, report generation)
- **Target users** (e.g., admin, lab staff)
- **Deployment environment** (e.g., LAN-based, desktop app)
- **Programming tools** used
- **Processes** that the system automates

Use verbs like:

- "allows"
 - "includes"
 - "supports"
 - "automates"
 - "enables"
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Part 2: Limitations (What the system does NOT cover)

Clearly describe:

- **Excluded technologies** (e.g., no RFID, no mobile access, no cloud)
- **User restrictions** (e.g., students cannot access)
- **Security/alerting gaps** (e.g., no overdue notifications)
- **Future features that are outside the current scope** (e.g., maintenance tracking, barcode printing)
- **Technical limitations** (e.g., works only on Windows, not yet cross-platform)

Use phrases like:

- "does not support"
 - "is limited to"
 - "is not designed for"
 - "does not include"
 - "is restricted to"
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FULLY EXPANDED SAMPLE CONTENT

F. Scope and Limitations

◆ Scope:

This study focuses on the development and implementation of a **Tools Inventory System** designed for the **BIT Automotive Program** of Iloilo Science and Technology University Dumangas Campus.

The system is developed as a **LAN-based application** that runs on a desktop environment using **PHP, MySQL, Apache, and Sublime Text** as the development stack. It allows only **authorized users**—specifically the administrator and laboratory staff—to access the system within the automotive laboratory.

The core features of the system include:

- Managing user accounts for lab staff and admin
- Maintaining a **tool and equipment registry** with descriptions and quantity
- Recording **stock-in transactions** (incoming tools/equipment)
- Recording **stock-out transactions** (tools removed or disposed)
- Recording **borrowing and returning of tools** by students (via staff entry)
- Generating **inventory summary reports** for print or review

All system activities are logged by user role and timestamp to maintain accountability. The system also allows basic printing of inventory reports and transaction history for documentation.

◆ **Limitations:**

While the developed system successfully automates key inventory functions, it has certain limitations due to time, budget, and technical scope:

1. **LAN-Only Access:** The system is restricted to use **within the laboratory** and cannot be accessed remotely or via the internet. There is **no web-based or cloud deployment** included in this version.
2. **No Student Login:** Students cannot access the system interface. All transactions (borrowing/returning) are handled by laboratory staff on their behalf.
3. **No RFID or Barcode Integration:** The system does not support **barcode scanning** or **RFID tracking** of tools. All transactions are manually encoded.
4. **No Notification or Alert System:** There are **no automatic notifications** for overdue borrowed tools or low stock levels.
5. **Platform Dependency:** The system is tested and optimized for **Windows OS only**. Compatibility with Linux or macOS is not guaranteed.
6. **No Maintenance or Calibration Module:** The system does not monitor tool maintenance schedules, calibration status, or tool conditions beyond basic remarks.
7. **Limited Reporting Options:** Reports generated are basic summaries and do not yet include advanced analytics like usage patterns or top-borrowed tools.